IV Infusions Practice 1

Module 3 - Intravenous Infusions · Daily practice sheet · No date

Name _____ Date _____

Continuous intravenous infusion of a potent analgesic is recommended for a patient with inoperable colon cancer and a body weight of 110 pounds. The dose is prepared by dissolving 200 mg of the drug in 500 mL of 5% dextrose. The drug is to be infused over 24 hours. The drug has a half-life of 4 hours and an apparent volume of distribution of 3 L/kg. Calculate the following:

a Rate of drug infusion in mg/hr

b Plasma concentration 6 hours after the start of the infusion

c Total amount of drug in the body 12 hours after the start of the infusion

d Steady-state plasma concentration

e Rate of drug infusion in mL/min

f An IV bolus loading dose

g Plasma concentration 12 hours after the cessation of the infusion